Data Semantics - Assignment 4 - NER and NEL

Riccardo Pozzi (riccardo.pozzi@unimib.it)

May 2026

General Information

This assignment is about Named Entity Recognition (NER) and Named Entity Linking (NEL), two building blocks of information extraction solutions centered around entities and supporting knowledge base population tasks.

You are provided with a Jupyter notebook[1] that contains the base code required for the exercises. The notebook includes the data you need (two example domain-specific documents) for the assignment and partial implementations.

You may choose to run the notebook in the cloud using Google Colab (suggested) or locally by installing Jupyter[2].

The assignment consists in completing the following instructions for each of the two example documents.

Exercise 1: NER with GliNER

- The two documents are domain specific and you are interested in finding specific entity types. These entity types are already given to you as `labels_cyber` and `labels_bio`.
- Apply GliNER to the two documents considering the respective labels.
- Visualize the annotated documents (e.g., with GateNLP).

Exercise 2: NEL

- Perform NEL with Wikipedia search engine (you can use the provided function `wikipedia_link`) on the entity mentions given by GliNER.

[1] `https://colab.research.google.com/drive/1HAWrxw59RZZXl6qkrts9YQbKNwMLAvxq?usp=sharing`

[2] `https://jupyter.org/install`

Exercise 3: Evaluate the results of NEL

- You find the correct entity set for each document in the variables `gold_cyber` and `gold_bio`.
- Follow the "entity tagging" formulation [1], and calculate *precision*, *recall*, and *F1 score* as described in the following example.
- The entity tagging formulation evaluates whether a NEL system correctly identifies that a document is about certain entities. This approach ignores the number of times each entity is mentioned and the positions of their mentions. Below is an example of evaluating entity linking using this formulation.

Entity Tagging Example

Example Text The entity mentions are in bold.

*"**Ronaldo** was born in **Brazil** and started his career at **Cruzeiro Esporte Clube**. Later, **Ronaldo** moved in **Barcelona**."*

This text is about the following entities: Ronaldo (Brazilian footballer), Brazil, Cruzeiro Esporte Clube, and Barcelona. So the expected set of entities (Gold Entities) G_E is:

$G_E = \{\text{Ronaldo (Brazilian footballer)}, \text{Brazil}, \text{Cruzeiro Esporte Clube}, \text{Barcelona}\};$

Example Prediction Our system missed *Barcelona* and *Brazil*, and misclassifying the sport club (Cruzeiro Esporte Clube) with the municipality Cruzeiro, São Paulo; the predicted entities P_E are as follows:

$P_E = \{\text{Ronaldo (Brazilian footballer)}, \text{Cruzeiro Esporte Clube}, \text{Cruzeiro São Paulo}\};$

In the entity tagging formulation we calculate *precision*, *recall*, and *F1 score* with the following formulas, where the operator $|X|$ is the cardinality of the set X and $A \cap B$ is the intersection of set A and B:

$$P = \frac{|P_E \cap G_E|}{|P_E|} \tag{1}$$

$$R = \frac{|P_E \cap G_E|}{|G_E|} \tag{2}$$

$$F1 = 2 * \frac{P * R}{P + R} \tag{3}$$

Thus, in this example:

$$|P_E \cap G_E| = |\{\text{Ronaldo (Brazilian footballer)}, \text{Cruzeiro Esporte Clube}\}| = 2$$

$$|P_E| = 3$$

$$|G_E| = 4$$

$$P = \frac{2}{3} = 0.67$$
$$R = \frac{2}{4} = 0.5$$
$$F1 = 0.57.$$

References

[1] Christina Du, Kashyap Popat, Louis Martin, and Fabio Petroni. Entity tagging: Extracting entities in text without mention supervision. arXiv:2209.06148, 2022.